FHIR Implementation for Lao PDR EIR

A guide for implementers and developers

Agenda

1. What is FHIR? (~10 min)

Resources, profiles, and why it matters

2. DHIS2 meets FHIR (~15 min)

How DHIS2 concepts map to FHIR, why Questionnaire

3. Defining FHIR metadata (~10 min)

Profiles, extensions, terminology using FSH

4. Building the integration (~10 min)

The pipeline from DHIS2 metadata to a FHIR API

5. FHIR for EIR in Lao PDR (~15 min)

CHR patients, organizations, IPS documents, live demo

Who is this for?

Implementers

- Know DHIS2 metadata (programs, data elements, option sets)
- Design profiles & terminology
- Coordinate standards adoption

Developers

- Build FHIR REST APIs
- Map DHIS2 data to FHIR resources
- Write FSH, validate & test

Both roles are needed – implementers define *what*, developers build *how*.

Part 1: What is FHIR?

What is FHIR?

Fast Healthcare Interoperability Resources – an HL7 standard.

- **REST + JSON** – same web standards as modern APIs
- **Resource-based** – everything is a Resource (Patient, Questionnaire, Organization…)
- **Profiling** – constrain base resources for your country/program
- **Open** – free specification, large ecosystem, no licensing

Standard: HL7 FHIR R4 (v4.0.1) – https://hl7.org/fhir/

Why FHIR?

Before FHIR

- Custom point-to-point integrations
- Different formats per system
- Expensive to build and maintain
- Breaks when systems change

With FHIR

- One standard API for all systems
- JSON/XML with defined structure
- Reusable profiles & terminology
- Large vendor/tool ecosystem

DHIS2 context: deployed in 80+ countries. FHIR provides a common language to exchange data between DHIS2, OpenMRS, lab systems, and national registries.

FHIR Resources We Need

Over 140 resource types exist – we use these for the Lao EIR:

Patient Demographics, identifiers, address

Questionnaire Form definitions (program stages)

QuestionnaireResponse Submitted form data (events)

Immunization *(IPS only)* Vaccine, date, dose, target disease

CodeSystem / ValueSet Coded vocabularies (option sets)

Organization / Location Org unit hierarchy, facility GPS coordinates

Bundle Search results, IPS documents

Resource Structure – Patient Example

A Lao CHR patient as FHIR JSON:

```
{
  "resourceType": "Patient",
  "id": "lao-patient-001",
  "meta": { "profile": ["http://moh.gov.la/fhir/StructureDefinition/chr-patient"] },
  "identifier": [{
    "system": "http://moh.gov.la/fhir/id/client-health-id",
    "value": "15032019-2-6092",
    "type": { "coding": [{ "code": "CHR", "display": "Community Health Record ID" }] }
  }],
  "name": [{ "family": "Phommasan", "given": ["Khamla"] }],
  "gender": "female",
  "birthDate": "2019-03-15"
}
```

Part 2: DHIS2 Meets FHIR

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DHIS2 → FHIR Mapping

DHIS2 concept	FHIR resource	Notes
Tracked Entity	Patient	Demographics + identifiers
Program Stage	Questionnaire	Form definition
Event	QuestionnaireResponse	Submitted form data
Data Element	Questionnaire item	Each field in the form
Option Set	ValueSet + CodeSystem	Coded answer options
Organisation Unit	Organization / Location	Facility hierarchy

Why Questionnaire?

FHIR has specific clinical resources (Encounter, Observation, Condition). Why not use those?

- **Simpler** – flat form structure, not deep clinical modeling
- **1:1 mapping** – Program Stage = Questionnaire, Event = Response
- **Familiar** – looks like a form/spreadsheet, not a clinical document
- **Linked to Patient** – each Response has a `subject` reference
- **Flexible** – any DHIS2 program can be represented without custom mapping
- **Trade-off** – less semantic richness, but much more practical for DHIS2 integration

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Questionnaire – DHIS2 Terms

DHIS2	FHIR Questionnaire	Example
Program Stage	Questionnaire	"ANC Visit"
Section	group item	"Vital Signs"
Data Element	item (with linkId)	"Systolic BP"
Data Element UID	linkId	`tVPKjkXrMSB`
Data Element type	item type	`integer`
Option Set	answerValueSet	Risk Factor options

A Questionnaire is the blank form template. It defines the questions but contains no answers.

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Questionnaire Structure

A form is a tree: **groups** (sections) contain **items** (questions):

```
ANC Visit Form
├── Visit Information (group)
│   ├── ANC visit #        — choice (→ ValueSet)  — linkId: kyclqodZVDs
│   └── Risk Factor        — choice (→ ValueSet)  — linkId: rxPKp45sFA2
├── Vital Signs (group)
│   ├── Systolic BP        — integer (mmHg)       — linkId: tVPKjkXrMSB
│   └── Fetus heart rate   — decimal (bpm)        — linkId: NjD9TNfcUu5
└── Lab Tests (group)
    ├── Hb test done?      — boolean              — linkId: uRM5be7E75w
    └── Hb result          — choice (→ ValueSet)  — linkId: FlkD0kQhHhJ
```

Each item has a **linkId** (data element UID), **text** (label), and **type**.

Item Types

Type	Input	DHIS2 equivalent
`string`	Free text	TEXT
`integer`	Whole number	INTEGER
`decimal`	Decimal number	NUMBER
`boolean`	Yes/No checkbox	BOOLEAN
`date`	Date picker	DATE
`choice`	Dropdown/radio	Option Set
`group`	Section container	Section

QuestionnaireResponse

A **submitted form with answers** – like a DHIS2 Event.

Contains

- **questionnaire** – which form was filled
- **subject** – which Patient
- **authored** – when submitted
- **status** – `completed`, `in-progress`
- **item** – the answers

Each answer has

- **linkId** – matches the question
- **answer** – typed value:
  - `valueInteger: 110`
  - `valueDecimal: 54.5`
  - `valueBoolean: true`
  - `valueDate: "2024-07-13"`
  - `valueCoding: { code, display }`

QuestionnaireResponse – Example

Khamla's ANC visit on June 15, 2024:

```
{
  "resourceType": "QuestionnaireResponse",
  "questionnaire": "http://moh.gov.la/fhir/Questionnaire/LaoANCVisit",
  "status": "completed",
  "subject": { "reference": "Patient/seed-patient-006" },
  "authored": "2024-06-15",
  "item": [
    { "linkId": "kyclqodZVDs", "text": "Number of ANC visit",
      "answer": [{ "valueCoding": { "code": "1", "display": "1st visit" } }] },
    { "linkId": "tVPKjkXrMSB", "text": "Systolic BP",
      "answer": [{ "valueInteger": 110 }] },
    { "linkId": "uRM5be7E75w", "text": "Hemoglobin test done?",
      "answer": [{ "valueBoolean": true }] }
  ]
}
```

How are they linked?

The Questionnaire **defines** the questions. The Response **captures** the data.

	Questionnaire (template)		QuestionnaireResponse (data)
linkId: **tVPKjkXrMSB**	Systolic BP, type: integer	→	answer: **110**
linkId: **rxPKp45sFA2**	Risk Factor, type: choice	→	answer: **#norisk**
linkId: **uRM5be7E75w**	Hb test done?, type: boolean	→	answer: **true**

The **linkId** is the same in both — that's how each answer is matched to its question.

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Part 3: Defining FHIR Metadata

FHIR Profiles

Profiles **constrain** base resources for local requirements:

```
Base Patient → DHIS2 Patient Profile → Lao CHR Patient Profile
```

- **Require** specific fields (e.g., CHR ID is mandatory)
- **Restrict** value sets (e.g., only Lao ethnicity codes)
- **Add extensions** for fields FHIR doesn't have
- **Slice identifiers** (DHIS2 UID, National ID, CHR ID, CVID)

Why Extensions?

FHIR's base Patient has name, gender, birthDate, address — but **not everything**:

CHR field	In base FHIR?	Solution
Name, gender, DOB	Yes	Standard fields
Province, district	Yes	`address.state`, `.district`
Ethnicity	No	**Extension**
Occupation	No	**Extension**
Blood group	No	**Extension**
Education level	No	**Extension**

Every country/program will need some extensions — this is normal and expected.

FHIR Terminology

Standardized vocabularies for coded data:

System	Use	Example
CVX	Vaccine codes	`19` = BCG, `94` = MR
SNOMED CT	Clinical terms	`56717001` = Tuberculosis
LOINC	Lab/observation codes	`29463-7` = Body weight
Local CodeSystems	Country-specific	Lao ethnicity, occupation

A **CodeSystem** defines the codes. A **ValueSet** picks which are valid for a specific field.

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Part 4: Building the Integration

Implementers & Developers

Implementers design

- Profiles – required fields, extensions
- Terminology – code systems for local concepts
- Identifier strategy – CHR ID, national IDs
- Workflow – search-or-create, EIR lookup
- Data quality – validation rules

Developers build

- FSH profiles from implementer designs
- Terminology from DHIS2 Option Sets
- Questionnaire definitions from Program Stages
- Validation with SUSHI + IG Publisher
- Integration layer (DHIS2 ↔ FHIR)

FSH – FHIR Shorthand

Human-readable syntax for authoring profiles:

```
Profile: LaoPatient
Parent: Patient
Title: "Lao PDR Patient"

* name 1..* MS          // at least one name, must support
* gender 1..1 MS        // exactly one gender, required
* birthDate 1..1 MS     // exactly one birth date, required
* address 0..* MS       // optional but must support
```

- `1..1` = required, `0..*` = optional, `1..*` = at least one
- `MS` = Must Support (client must handle this field)

FSH – Extensions & Terminology

```
Extension: ChrEthnicity
Title: "Ethnicity"
* value[x] only CodeableConcept
* valueCodeableConcept from ChrEthnicityVS (required)

CodeSystem: ChrEthnicityCS
* #lao "Lao"
* #khmou "Khmou"
* #hmong "Hmong"
* #phouthai "Phouthai"

ValueSet: ChrEthnicityVS
Title: "Lao Ethnicity"
* include codes from system ChrEthnicityCS
```

FSH – Instances (Examples)

```
Instance: ExampleLaoPatient
InstanceOf: LaoPatient
Title: "Example – Khamla Phommasan"
Usage: #example

* name[0].family = "Phommasan"
* name[0].given[0] = "Khamla"
* gender = #female
* birthDate = "2019-03-15"
* address[0].city = "Anou"
* address[0].state = "Vientiane Capital"
* address[0].country = "LA"
* extension[ethnicity].valueCodeableConcept
    = ChrEthnicityCS#lao "Lao"
```

Examples are included in the IG and used for validation testing.

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Tooling Overview

Tool	Purpose
SUSHI	Compile FSH → FHIR JSON
IG Publisher	Build full Implementation Guide (HTML)
HAPI FHIR	Java-based FHIR server
Postman	Test FHIR REST APIs
FastAPI	Python web framework (this project)

Part 5: FHIR for EIR in Lao PDR

Lao CHR Overview

Community Health Record – national health information system built on DHIS2.

- **Patient registry** with 6+ identifier types
- **EIR** – Electronic Immunization Registry (national EPI schedule)
- **FHIR mapping** – CHR data exposed as FHIR resources
- **Search** – by identifier, name, demographics, phone, village

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CHR Patient – Identifiers

Each patient can have multiple identifiers across systems:

Type	FHIR System	Example
CHR ID	`client-health-id`	`15032019-2-6092`
National ID	`green-national-id`	`LA-1998-41270`
CVID	`cvid`	`CVID-60104292`
Passport	`passport`	`LA-P482917`
Insurance	`insurance`	`INS-LA-30421`
Family Book	`family-book`	`FB-60293`

In DHIS2: each is a Tracked Entity Attribute. In FHIR: each is a slice on `Patient.identifier`.

CHR Patient – Extensions & Address

Custom extensions

- Nationality
- Ethnicity (49 ethnic groups)
- Occupation
- Education level
- Blood group

Address mapping

CHR field	FHIR field
Province	`address.state`
District	`address.district`
Village	`address.city`

EIR – Lao EPI Schedule

Vaccines mapped to FHIR Immunization using **CVX codes**:

Vaccine	CVX	Age
BCG	19	Birth
Hepatitis B (birth)	45	Birth
OPV (3 doses)	2	2, 4, 6 mo
Pentavalent (3 doses)	102	2, 4, 6 mo
PCV13 (3 doses)	152	2, 4, 6 mo
Measles-Rubella (2 doses)	94	9, 18 mo

EIR – Immunization Resource

```
{
  "resourceType": "Immunization",
  "status": "completed",
  "patient": { "reference": "Patient/seed-patient-006" },
  "vaccineCode": {
    "coding": [{ "system": "http://hl7.org/fhir/sid/cvx",
                 "code": "19", "display": "BCG" }]
  },
  "occurrenceDateTime": "2019-03-15",
  "protocolApplied": [{
    "doseNumberPositiveInt": 1,
    "targetDisease": [{
      "coding": [{ "system": "http://snomed.info/sct",
                   "code": "56717001", "display": "Tuberculosis" }]
    }]
  }]
}
```

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International Patient Summary

IPS = cross-border immunization verification.

- A **document Bundle** containing:
  - **Composition** – table of contents
  - **Patient** – demographics
  - **Immunizations** – vaccination history
- Critical for Lao PDR with 4 neighboring countries
- Standardized format – any IPS-compliant system can read it

IPS – Bundle Structure

```
{
  "resourceType": "Bundle",
  "type": "document",
  "entry": [
    { "resource": { "resourceType": "Composition",
                    "title": "International Patient Summary" } },
    { "resource": { "resourceType": "Patient",
                    "name": [{"family": "Vongsa", "given": ["Somchai"]}] } },
    { "resource": { "resourceType": "Immunization",
                    "vaccineCode": {"coding": [{"code": "19", "display": "BCG"}]} } },
    { "resource": { "resourceType": "Immunization",
                    "vaccineCode": {"coding": [{"code": "94", "display": "MR"}]} } }
  ]
}
```

Live Demo

http://localhost:8000

Patient Search – browse, filter, identifier lookup

CHR Dashboard – stats, search, registration

EIR – immunization history per patient

IPS Bundles – document view with composition

Questionnaires – fill & submit QuestionnaireResponses

Organizations – DHIS2 org unit hierarchy & facility locations

Terminology – browse ValueSets & CodeSystems, $expand

FHIR API – Patient, Questionnaire, ValueSet/$expand, Postman collection

Hands-on Exercises

Exercises 1–5:

#	Topic	What you'll build
01	Profiles	Constrain Patient for DHIS2
02	Extensions	Add Organisation Unit
03	Value Sets & Code Systems	Define coded vocabularies
04	Instances	Create example data
05	Logical Models	Model a DHIS2 Tracked Entity

Each has `instructions.md`, `starter.fsh`, and `solution.fsh`. Compile with `make docker-sushi`.

Exercises 6–10

#	Topic	What you'll build
06	Mappings	Map Tracked Entity → Patient
07	RuleSets	Reusable rule patterns
08	Aliases	URI shorthand
09	Invariants	FHIRPath validation rules
10	Slicing	Split identifiers (DHIS2 UID vs National ID)

Summary

What we covered

- **FHIR basics** – resources, JSON structure, REST API
- **DHIS2 ↔ FHIR mapping** – why Questionnaire, how linkId works
- **Profiles & terminology** – constraining resources, extensions, ValueSets
- **The pipeline** – DHIS2 metadata → FSH → SUSHI → FHIR API
- **Lao EIR** – CHR patients, organizations, IPS cross-border

Try It Yourself

Run the app

```
make seed    # generate test data
make run     # http://localhost:8000
```

Do the exercises

```
exercises/01-profiles/instructions.md
```

Read the docs

```
make docs-serve
```

Resources & Questions

Links

- **FHIR spec:** https://hl7.org/fhir/
- **FSH/SUSHI:** https://fshschool.org/
- **DHIS2 + FHIR:** https://dhis2.org/integration/fhir/

Questions?